PCD Homework 1 Report

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1 Experiment Description

This experiment evaluates the performance and reliability of five different transport protocol implementations:

- **TCP**: Standard stream-oriented reliable transport.
- **UDP**: Raw unreliable datagram transport.
- **UDP (Stop and Wait)**: A custom reliability layer implemented over UDP requiring an ACK for every packet.
- **QUIC**: Used in Datagram mode to allow for high-throughput and to allow for testing the protocol's data loss without the built in reliability of stream mode.
- **QUIC (Stop and Wait)**: A custom reliability layer implemented over QUIC datagrams.

The metrics were collected by transferring two different payload sizes (500 MB and 1 GB) using varying block sizes (512 B, 1KB, 16KB, 60KB) over a local loopback connection. Quic does not support datagram sizes larger than 1.5KB, so the experiments were run with 512 B and 1KB.

2 How to Run

The project requires Rust and Cargo to be installed. The server and client can be run using the following commands:

2.1 Running the Server

The server listens to traffic for all the protocols concurrently on separate ports. To start the server, run the following command:

```
cargo run -p server
```

2.2 Running the Client

```
cargo run -p client -- --transport <TYPE> --file-path <PATH> --block-size <BYTES>
```

2.2.1 Client Options

- **–transport TYPE**: The transport protocol to use. Can be **tcp**, **udp**, **udp-stop-and-wait**, **quic**, or **quic-stop-and-wait**.
- **–file-path PATH**: The path to the file to transfer.
- **–block-size BYTES**: The size of the block to transfer.

Note: When using UDP with larger block sizes, the UDP maximum datagram size of the system may need to be increased.

```
sudo sysctl -w net.inet.udp.maxdgram=65535
```

3 Experiment Results

3.1 TCP

Payload	Block	Time(s)	Pkts(S)	Pkts(R)	Bytes(S)	Bytes(R)	Loss%
1 GB	512	19.6306	2097152	2097152	1073741824	1073741824	0.00%
1 GB	1024	9.6824	1048576	1048576	1073741824	1073741824	0.00%
1 GB	16384	1.2510	65536	65536	1073741824	1073741824	0.00%
1 GB	60000	0.5400	17896	17896	1073741824	1073741824	0.00%
500 MB	512	10.4824	1024000	1024000	524288000	524288000	0.00%
500 MB	1024	5.0572	512000	512000	524288000	524288000	0.00%
500 MB	16384	0.6705	32000	32000	524288000	524288000	0.00%
500 MB	60000	0.2905	8739	8739	524288000	524288000	0.00%

3.2 UDP

Payload	Block	Time(s)	Pkts(S)	Pkts(R)	Bytes(S)	Bytes(R)	Loss%
1 GB	512	20.0796	2097152	2094517	1073741824	1072392704	0.13%
1 GB	1024	10.1761	1048576	1046042	1073741824	1071147008	0.24%
1 GB	16384	0.7629	65536	62467	1073741824	1023459328	4.68%
1 GB	60000	0.3009	17896	15836	1073741824	950141824	11.51%
500 MB	512	9.8073	1024000	1021243	524288000	522876416	0.27%
500 MB	1024	4.9123	512000	509365	524288000	521589760	0.51%
500 MB	16384	0.3790	32000	28771	524288000	471384064	10.09%
500 MB	60000	0.1666	8739	7062	524288000	423668000	19.19%

3.3 UDP (Stop and Wait)

Payload	Block	Time(s)	Pkts(S)	Pkts(R)	Bytes(S)	Bytes(R)	Loss%
1 GB	512	110.0781	2097152	2097152	1073741824	1073741824	0.00%
1 GB	1024	55.2263	1048576	1048576	1073741824	1073741824	0.00%
1 GB	16384	3.6949	65536	65536	1073741824	1073741824	0.00%
1 GB	60000	1.0859	17896	17896	1073741824	1073741824	0.00%
500 MB	512	53.7394	1024000	1024000	524288000	524288000	0.00%
500 MB	1024	26.9609	512000	512000	524288000	524288000	0.00%
500 MB	16384	1.7910	32000	32000	524288000	524288000	0.00%
500 MB	60000	0.5494	8739	8739	524288000	524288000	0.00%

3.4 QUIC

Payload	Block	Time(s)	Pkts(S)	Pkts(R)	Bytes(S)	Bytes(R)	Loss%
1 GB	512	14.6380	2097152	2041553	1073741824	1045275136	2.65%
1 GB	1024	13.4797	1048576	989332	1073741824	1013075968	5.65%
500 MB	512	6.7652	1024000	973617	524288000	498491904	4.92%
500 MB	1024	5.9717	512000	447697	524288000	458441728	12.56%

3.5 QUIC (Stop and Wait)

Payload	Block	Time(s)	Pkts(S)	Pkts(R)	Bytes(S)	Bytes(R)	Loss%
1 GB	512	151.8875	2097152	2097152	1073741824	1073741824	0.00%
1 GB	1024	75.7086	1048576	1048576	1073741824	1073741824	0.00%
500 MB	512	74.0261	1024000	1024000	524288000	524288000	0.00%
500 MB	1024	37.0389	512000	512000	524288000	524288000	0.00%

4 Visualized Results

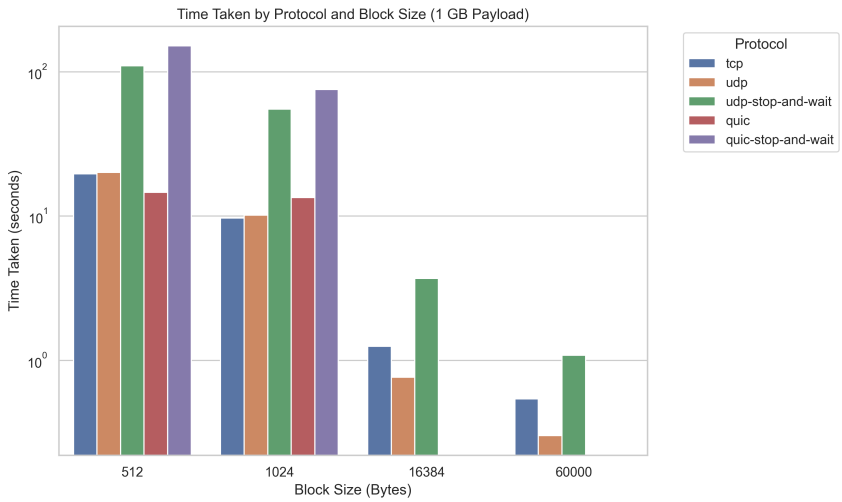

Figure 1: Time taken by protocol and block size (1 GB payload).

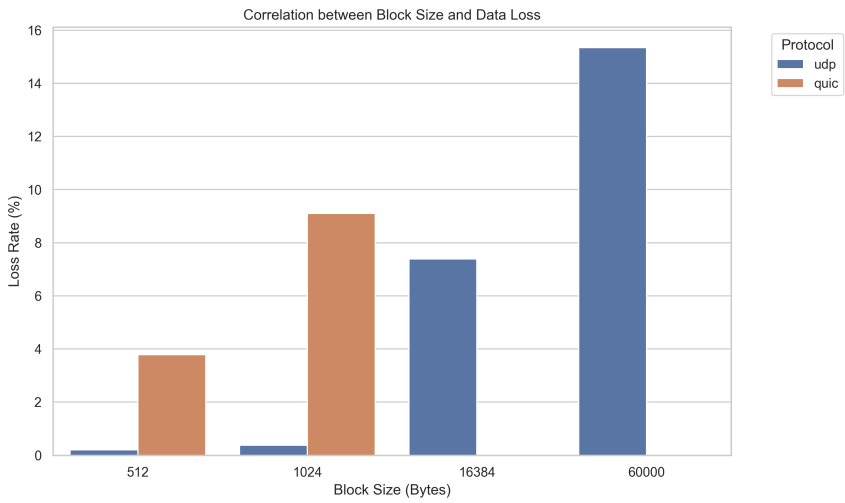

Figure 2: Correlation between block size and data loss for unreliable protocols (UDP and QUIC).

5 Results Analysis

- **Throughput Efficiency**: For all protocols, increasing the block size resulted in a massive reduction in transfer time due to lower per-packet overhead and fewer system calls.
- **Reliability vs. Latency**: The "Stop and Wait" implementations achieved 100% reliability but were significantly slower as they wait for an acknowledgement for every individual packet.
- **Block Size vs. Data Loss**: In unreliable transports (UDP and QUIC), there is a direct correlation between block size and the probability of data loss. Larger blocks (e.g., 60,000 B) force the underlying layers to use IP fragmentation (in UDP). Because datagrams are atomic, the loss of a single byte or fragment results in the discarding of the entire block. Consequently, while larger blocks improve throughput by reducing header-to-payload ratios, they significantly increase the loss risk, leading to the observed 11–19% loss rates compared to the sub-1% loss seen with smaller blocks.
- **Security Overhead**: The mandatory TLS 1.3 encryption in QUIC introduces a measurable performance "tax" compared to raw UDP. Each datagram requires CPU-intensive cryptographic operations for packet protection.

- **Time Efficiency**: Raw UDP was the fastest overall for large blocks (e.g., 0.30s for 1 GB), as it has the lowest per-packet overhead and no congestion control. TCP followed closely (0.54s for 1 GB) due to its highly optimized OS-level stack. Interestingly, at small block sizes (512 B), QUIC (14.64s) was more efficient than both TCP (19.63s) and UDP (20.08s), demonstrating QUIC's superior handling of high-packet-rate streams and its reduced handshake overhead for the first few packets.

6 Conclusions

The experiments demonstrate that there is no "one-size-fits-all" protocol; the choice depends heavily on the specific requirements for reliability, latency, and security.

6.1 TCP (Transmission Control Protocol)

- **Advantages**: Guaranteed 100% reliability and ordering. Highly optimized OS-level implementation. No data loss regardless of block size.
- **Disadvantages**: Vulnerable to "Head-of-Line Blocking". Higher overhead due to heavy handshake and congestion control.

6.2 UDP (User Datagram Protocol)

- **Advantages**: Minimal overhead and lowest possible latency. Maximum theoretical throughput for large block sizes.
- **Disadvantages**: No reliability or ordering guarantees. High packet loss for large blocks due to IP fragmentation sensitivity.

6.3 QUIC (Quick UDP Internet Connections)

- **Advantages**: Built-in TLS 1.3 security. Better handling of high-packet-rate streams (small blocks). Supports connection migration.
- **Disadvantages**: Measurable CPU overhead from mandatory encryption. Datagram mode is strictly MTU-limited and lacks native fragmentation.